Continuous Optimisation - I

Functions of several variables and partial differentiation

Find the gradient of following functions at the given point as a column vector

1. $u = e^{xyz}$ and $P = (1,1,1)$

partially diff u w r t x, y, z respectively

$u_x = yz e^{xyz}$, $u_y = xz e^{xyz}$, $u_z = xy e^{xyz}$

Gradient $\nabla u = \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} = \begin{pmatrix} yz e^{xyz} \\ xz e^{xyz} \\ xy e^{xyz} \end{pmatrix}$

at $P = (1,1,1) \Rightarrow \nabla u(1,1,1) = \begin{pmatrix} e \\ e \\ e \end{pmatrix}$

2. $f = xyz$ and $P = (1,0,1)$

$f_x = yz$, $f_y = xz$ and $f_z = xy$

$\nabla f = \begin{bmatrix} yz \\ xz \\ xy \end{bmatrix} \Rightarrow \nabla f(1,0,1) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

3. $g = x^2y^2z$ and $P = (2,3,-1)$

$g_x = 2xy^2z$, $g_y = 2x^2yz$, $g_z = x^2y^2$, $\nabla g = \begin{bmatrix} 2xy^2z \\ 2x^2yz \\ x^2y^2 \end{bmatrix} \Rightarrow \nabla g(2,3,-1) = \begin{pmatrix} -36 \\ -24 \\ 36 \end{pmatrix}$

4. $g = 3x^2 + y^2 + z^2$ and $P = (0,0,2)$

$g_x = 6x$, $g_y = 2y$, $g_z = 2z$, $\nabla g = \begin{pmatrix} 6x \\ 2y \\ 2z \end{pmatrix} \Rightarrow \nabla g(0,0,2) = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$

5. $g = 3xe^{2y} + e^z y^2 + z^2$ and $P = (1,0,0)$

$g_x = 3e^{2y}$, $g_y = 6xe^{2y} + 2e^z y$, $g_z = e^z y^2 + 2z$

$\nabla g = \begin{pmatrix} 3e^{2y} \\ 6xe^{2y} + 2e^z y \\ e^z y^2 + 2z \end{pmatrix} \Rightarrow \nabla g(1,0,0) = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$

6. $f = 2x^3 e^{y^2} z^2$ and $P = (2,0,3)$

$f_x = 6x^2 e^{y^2} z^2$, $f_y = 4x^3 e^{y^2} y z^2$, $f_z = 4x^3 e^{y^2} z$

$\nabla f = \begin{pmatrix} 6x^2 e^{y^2} z^2 \\ 4x^3 e^{y^2} y z^2 \\ 4x^3 e^{y^2} z \end{pmatrix} \Rightarrow \nabla f(2,0,3) = \begin{pmatrix} 216 \\ 0 \\ 96 \end{pmatrix}$

Jacobian Matrix: The Jacobian matrix is a matrix composed of the first order partial derivatives of a multivariable function.

Let $f: \mathbb{R}^n \to \mathbb{R}^m$ be defined by $f(x_1, x_2, \ldots x_n) = (f_1, f_2, \ldots f_m)$ then Jacobian of f is defined as

$$J_x(f) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & & & \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{pmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \\ \vdots \\ \frac{df_n}{dx} \end{pmatrix} = \frac{df}{dx} = Df$$

how each output function changes wrt each input variable.

Find the Jacobian matrix of the following function

1. $f(x,y) = (x^4 + 3y^2x,\ 5y^2 - 2xy + 1)$ at $(1,2)$

Let $f_1 = x^4 + 3y^2x$, and $f_2 = 5y^2 - 2xy + 1$

$f_{1x} = \frac{\partial f_1}{\partial x} = 4x^3 + 3y^2$ and $f_{1y} = \frac{\partial f_1}{\partial y} = 6yx$

$f_{2x} = \frac{\partial f_2}{\partial x} = -2y$ $\qquad f_{2y} = \frac{\partial f_2}{\partial y} = 10y - 2x$

$$\therefore J = \begin{bmatrix} 4x^3+3y^2 & 6yx \\ -2y & 10y-2x \end{bmatrix}$$

at $(1,2) \Rightarrow J(1,2) = \begin{bmatrix} 16 & 12 \\ -4 & 18 \end{bmatrix}$

2. $f(x,y) = (e^{xy} + y,\ y^2x)$ at $(0,-2)$.

Let $f_1 = e^{xy} + y$, $f_2 = y^2x$

$$J = \begin{bmatrix} f_{1x} & f_{1y} \\ f_{2x} & f_{2y} \end{bmatrix} = \begin{bmatrix} ye^{xy} & xe^{xy}+1 \\ y^2 & 2xy \end{bmatrix} \Rightarrow J_{(0,-2)} = \begin{bmatrix} -2 & 1 \\ 4 & 0 \end{bmatrix}$$

3. $f(x,y) = (x^3y^2 - 5x^2y^2,\ y^6 - 3y^3x + 7)$ at $(1,-1)$

Let $f_1 = x^3y^2 - 5x^2y^2$, $f_2 = y^6 - 3y^3x + 7$

$$J = \begin{bmatrix} f_{1x} & f_{1y} \\ f_{2x} & f_{2y} \end{bmatrix} = \begin{bmatrix} 3x^2y^2 - 10xy^2 & 2x^3y - 10x^2y \\ -3y^3 & 6y^5 - 9y^2x \end{bmatrix}$$

$$J(1,-1) = \begin{bmatrix} -7 & 8 \\ 3 & -15 \end{bmatrix}$$

4. $f(x,y,z) = (z\tan(x^2-y^2),\ xy\ln(z/2))$ at $(2,-2,2)$

Let $f_1 = z\tan(x^2-y^2)$ and $f_2 = xy\ln(z/2)$

$$J = \begin{bmatrix} f_{1x} & f_{1y} & f_{1z} \\ f_{2x} & f_{2y} & f_{2z} \end{bmatrix} = \begin{bmatrix} z\sec^2(x^2-y^2).2x & z\sec^2(x^2-y^2)(-2y) & \tan(x^2-y^2) \\ y\ln(z/2) & x\ln(z/2) & xy\,\frac{1}{z} \end{bmatrix}$$

$$J(2,-2,2) = \begin{bmatrix} 8 & 8 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

5. $f(x,y,z) = (xe^{2y}\cos(-z),\ (y-2)^3\sin(z/2),\ e^{2y}\ln(x/3))$ at $(3,0,\pi)$

Let $f_1 = xe^{2y}\cos z$, $f_2 = (y-2)^3\sin(z/2)$, $f_3 = e^{2y}\ln(x/3)$

$$J = \begin{bmatrix} f_{1x} & f_{1y} & f_{1z} \\ f_{2x} & f_{2y} & f_{2z} \\ f_{3x} & f_{3y} & f_{3z} \end{bmatrix} = \begin{bmatrix} e^{2y}\cos z & 2xe^{2y}\cos z & -xe^{2y}\sin z \\ 0 & 3(y-2)^2\sin z/2 & (y-2)^3/2\ \cos(z/2) \\ e^{2y}\frac{1}{x} & 2e^{2y}\ln(x/3) & 0 \end{bmatrix}$$

$$J_{(3,0,\pi)} = \begin{bmatrix} -1 & -6 & 0 \\ 0 & 12 & 0 \\ 1/3 & 0 & 0 \end{bmatrix}$$

Local and Global Optima :

Def^n: Let $X \subset \mathbb{R}^2$, Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function, and let $P \in X$.

i) f has a local maximum at P if there exist neighbourhood U of P in $X \ni f(P) \geq f(x)$ for all $x \in U$.

ii) f has a local minimum at P $\exists$ neighbourhood U of P in $X \ni$ $f(P) \leq f(x)\ \forall\ x \in U$.

iii) f has a global maximum at P if $f(P) \geq f(x)\ \forall\ x \in U$, in which case we call $f(P)$ the global maximum value of f.

iv) f has a global minimum at P if $f(P) \leq f(x)\ \forall x \in U$, in which case we call $f(P)$ the global minimum value of f.

Second derivative test : Let $z = f(x,y)$ be the differentiable function. For a critical point (a,b), let

$$D = f_{xx}(a,b)\,f_{yy}(a,b) - (f_{xy}(a,b))^2$$ be the discriminant.

* If $D > 0$ and $f_{xx}(a,b) > 0$, then f has a local minimum at (a,b).
* If $D > 0$ and $f_{xx}(a,b) < 0$, then f has a local maximum at (a,b).
* If $D < 0$, then (a,b) is the saddle point of f.

* If $D = 0$, then the test gives no information.

To find optimum point of multivariate functions more than two variables, we need to compute its Hessian matrix at all the critical points. Hessian matrix of function of three variables $f(x, y, z)$ is given by

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$$

Second derivative test : Find the critical points, plug them in the Hessian matrix, and compute their eigenvalues.

* If all the eigenvalues are strictly positive, then the critical point is a local minimum.
* If all the eigenvalues are stricty negative, then the critical point is a local maximum.
* If H has both positive & negative eigenvalues and all are non-zero, then the critical point is saddle point.
* If any of the eigenvalue is zero, then the test gives no information.

For the following multivariate function

i) Find all the stationary points of the function

ii) Find the Hessian matrix

iii) Classify the stationary points.

ⓐ $f(x, y) = x^2 + y^2 - 4x + 2y$

Soln: $f_x = 2x - 4$, $f_y = 2y + 2$

$f_x = 0$ & $f_y = 0$

$2x - 4 = 0$ $\quad$ $2y + 2 = 0$

$x = 2$ $\quad$ $y = -1$

$\therefore$ The stationary point is $(2, -1)$

The Hessian matrix is $H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$D = f_{xx} f_{yy} - (f_{xy})^2 = 4 - 0 = 4 > 0$

$D > 0$, $f_{xx} = 2 > 0$

$\therefore$ The function $f(x, y) = x^2 + y^2 - 4x + 2y$ local minimum at $(2, -1)$.

2. $f(x,y,z) = xy + yz + zx - 4x + 2y$

Solⁿ: $f_x = y + z - 4$, $f_y = x + z + 2$, $f_z = x + y$

$\Rightarrow y + z - 4 = 0$, $x + z + 2 = 0$, $x + y = 0 \Rightarrow x = -y$

$\Rightarrow -x + z = 4$ $\quad \Rightarrow x + 4 + x + 2 = 0$ $\quad \therefore y = 3$ and

$z = 4 + x$ $\quad 2x = -6$ $\quad z = 1$

$\boxed{x = -3}$

$\therefore$ The stationary point is $(-3, 3, 1)$.

The Hessian matrix $H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$\therefore D = f_{xx} f_{yy} - (f_{xy})^2 = 0 - 1$

From Hessian matrix $\lambda^3 - 3\lambda - 2 = 0 \Rightarrow \lambda = 2, -1, -1$

H has both positive & negative eigenvalues

$\therefore (-3, 3, 1)$ is a saddle point.

3. $f(x,y,z) = x^2 + y^2 + z^2 + xy + yz + zx$

Solⁿ: $f_x = 2x + y + z$, $f_y = 2y + x + z$, $f_z = 2z + y + x$

$2x + y + z = 0$ — ①, $2y + x + z = 0$ — ②, $2z + y + x = 0$ — ③

From ① & ②

$2x + y + z = 0$

$x + 2y + z = 0$

$x - y = 0 \Rightarrow x = y$

$\therefore$ ① becomes $3x + z = 0 \Rightarrow z = -3x$

$D = 4 - 1 = 3 > 0$

③ $\Rightarrow -6x + 2x = 0 \Rightarrow x = 0$, $y = 0$, $z = 0$

$\therefore$ The stationary point is $(0, 0, 0)$.

$H = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ $\Rightarrow \lambda^3 - 6\lambda^2 + \{(4-1) + (4-1) + (4-1)\}\lambda - 4 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$

$\Rightarrow \lambda = 4, 1, 1$

all eigen values are strictly positive

$\therefore f$ is local minimum at $(0, 0, 0)$.

4. $f(x,y,z) = -x^3 + 3xz + 2y - y^2 - 3z^2$

Sol^n : $f_x = -3x^2 + 3z$, $f_y = 2 - 2y$, $f_z = 3x - 6z$

$-3x^2 + 3z = 0$, $2 - 2y = 0 \Rightarrow y = 1$, $3x - 6z = 0$

$-12z^2 + 3z = 0$ $\Rightarrow 3x = 6z \Rightarrow x = 2z$

$z(3 - 12z) = 0$

$\Rightarrow z = 0$ & $3 - 12z = 0 \Rightarrow z = 1/4$

If $z = 0 \Rightarrow x = 0$ and $z = 1/4 \Rightarrow x = 1/2$

$\therefore$ The stationary points are $(0, 1, 0)$, $(1/2, 1, 1/4)$

$$H = \begin{bmatrix} -6x & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{bmatrix}$$

at $(0,1,0)$ $H = \begin{bmatrix} 0 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{bmatrix}$ $\lambda^3 + 8\lambda^2 + (12 + (-9) + 0)\lambda - 18 = 0$

$\lambda = -2, -7.24, 1.24$

H has both +ve & -ve

$\therefore$ $(0, 1, 0)$ is a saddle point.

at $(1/2, 1, 1/4)$ $H = \begin{bmatrix} -3 & 0 & 3 \\ 0 & -2 & 0 \\ 3 & 0 & -6 \end{bmatrix}$ $\lambda^3 + 11\lambda^2 + (12 + (18 - 9) + 6)\lambda + 18 = 0$

$\Rightarrow \lambda = -7.85, -1.14, -2$

all eigen values are negative.

5. $f(x, y, z) = x^2 + y^2 + z^2$

$f_x = 2x$, $f_y = 2y$, $f_z = 2z$

$x = 0$, $y = 0$, $z = 0$

$\therefore$ $(0, 0, 0)$ is a stationary point

$H = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ $\lambda^3 - 6\lambda^2 + (4 + 4 + 4)\lambda - 8 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$

$\lambda = 2, 2, 2$

$\therefore$ Eigen values are positive

Hence ~~the~~ f is local minimum at $(0, 0, 0)$

1. Discuss the gradient of vector with respect to matrix.

Let $f = (f_1\ f_2 \dots f_m)$ be a vector and $X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & & & \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}$

$$\frac{df}{dX} = \begin{bmatrix} \frac{df_1}{dX} & \frac{df_2}{dX} & \dots & \frac{df_m}{dX} \end{bmatrix}$$

$$= \left[\begin{bmatrix} \frac{\partial f_1}{\partial x_{11}} & \frac{\partial f_1}{\partial x_{12}} & \dots & \frac{\partial f_1}{\partial x_{1n}} \\ \frac{\partial f_1}{\partial x_{21}} & \frac{\partial f_1}{\partial x_{22}} & \dots & \frac{\partial f_1}{\partial x_{2n}} \\ \vdots & & & \\ \frac{\partial f_1}{\partial x_{m1}} & \frac{\partial f_1}{\partial x_{m2}} & \dots & \frac{\partial f_1}{\partial x_{mn}} \end{bmatrix} \begin{bmatrix} \frac{\partial f_2}{\partial x_{11}} & \frac{\partial f_2}{\partial x_{12}} & \dots & \frac{\partial f_2}{\partial x_{1n}} \\ \vdots & & & \\ \frac{\partial f_2}{\partial x_{m1}} & \frac{\partial f_2}{\partial x_{m2}} & \dots & \frac{\partial f_2}{\partial x_{mn}} \end{bmatrix} \dots \begin{bmatrix} \frac{\partial f_m}{\partial x_{11}} & \frac{\partial f_m}{\partial x_{12}} & \dots & \frac{\partial f_m}{\partial x_{1n}} \\ \vdots & & & \\ \frac{\partial f_m}{\partial x_{m1}} & \dots & & \frac{\partial f_m}{\partial x_{mn}} \end{bmatrix} \right]$$

2. Obtain the gradient of vector $f = [e^{x_0 x_1}\ e^{x_2 x_3}]$ wrt the matrix

$x = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$

$$\frac{df}{dX} = \begin{bmatrix} \frac{df_1}{dX} & \frac{df_2}{dX} \end{bmatrix} = \left[\begin{bmatrix} \partial f_1/\partial x_0 & \partial f_1/\partial x_1 \\ x_1 e^{x_0 x_1} & x_0 e^{x_0 x_1} \\ \partial f_1/\partial x_2 & \partial f_1/\partial x_3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \partial f_2/\partial x_0 & \partial f_2/\partial x_1 \\ 0 & \partial f_2/\partial x_3 \\ \partial f_2/\partial x_2 & \\ x_3 e^{x_2 x_3} & x_2 e^{x_2 x_3} \end{bmatrix} \right]$$

3. Obtain the gradient of scalar $\phi = 4x_0 + 2x_1 - 3x_2 + x_4$ wrt matrix

$x = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$

Sol^n: $$\frac{d\phi}{dX} = \begin{bmatrix} \partial\phi/\partial x_0 & \partial\phi/\partial x_1 \\ \partial\phi/\partial x_2 & \partial\phi/\partial x_3 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ -3 & 1 \end{bmatrix}$$

4. Find the derivative of the vector $\begin{bmatrix} x_0 x_1 \\ x_0 + x_1 \\ x_2 \end{bmatrix}$ wrt vector $[x_0\ x_1\ x_2]$ & hence find the nature of the resulting matrix at $(0,1,1)$

$$\frac{df}{dx} = \begin{bmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \\ \frac{df_3}{dx} \end{bmatrix} = \begin{bmatrix} [\partial f_1/\partial x_0 & \partial f_1/\partial x_1 & \partial f_1/\partial x_2] \\ [\partial f_2/\partial x_0 & \partial f_2/\partial x_1 & \partial f_2/\partial x_2] \\ [\partial f_3/\partial x_0 & \partial f_3/\partial x_1 & \partial f_3/\partial x_2] \end{bmatrix} = \begin{bmatrix} [x_1 & x_0 & 0] \\ [1 & 1 & 0] \\ [0 & 0 & 1] \end{bmatrix}$$

at $(0,1,1)$ $\frac{df}{dx}(0,1,1) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\lambda^3 - 3\lambda^2 + (1+1+1)\lambda - 1 = 0 \Rightarrow \lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$

$\lambda = 1, 1, 1$ $\quad\therefore$ all eigenvalues are strictly +ve

$\therefore$ Nature is positive definite

5. Obtain the gradient of the vector $\begin{bmatrix} 4x_0x_1 \\ 4x_0+2x_1 \\ 3x_2 \end{bmatrix}$ wrt vector $[x_0\ x_1\ x_2]$ and hence find the values of x_0 & x_1 for which 8, 3 & 2 are the eigenvalues of derivative matrix

$$\frac{df}{dx} = \begin{bmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \\ \frac{df_3}{dx} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_0} & \frac{\partial f_1}{\partial x_1} & \partial f_1/\partial x_2 \\ \frac{\partial f_2}{\partial x_0} & \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \\ \partial f_3/\partial x_0 & \partial f_3/\partial x_1 & \partial f_3/\partial x_2 \end{bmatrix} = \begin{bmatrix} 4x_1 & 4x_0 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Given eigenvalues are 8, 3, 2.

WKT Trace = Sum of the eigenvalues

$4x_1+2+3 = 8+3+2 \Rightarrow 4x_1+5=13 \Rightarrow x_1 = 2$

WKT product of eigenvalues is equal to determinant

$\therefore |A| = 16$ — (i)

Consider 2×2 matrix $\begin{bmatrix} 8 & 4x_0 \\ 4 & 2 \end{bmatrix}$

$|A| = 16-16x_0 \Rightarrow 16 = 16-16x_0 \Rightarrow x_0 = 0$ | from (i).

$$\therefore \frac{df}{dx} = \begin{bmatrix} 8 & 0 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

6. Obtain the gradient of the matrix $\begin{bmatrix} x_0x_1 \\ x_0+e^{x_1} \\ x_2^2 \end{bmatrix}_{3\times1}$ wrt $\begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}_{3\times1}$

Sol^n:
$$\frac{df}{dx} = \begin{bmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \\ \frac{df_3}{dx} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \partial f_1/\partial x_0 \\ \partial f_1/\partial x_1 \\ \partial f_1/\partial x_2 \end{bmatrix} \\ \begin{bmatrix} \partial f_2/\partial x_0 \\ \partial f_2/\partial x_1 \\ \partial f_2/\partial x_2 \end{bmatrix} \\ \begin{bmatrix} \partial f_3/\partial x_0 \\ \partial f_3/\partial x_1 \\ \partial f_3/\partial x_2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} x_1 \\ x_0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 1 \\ e^{x_1} \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 2x_2 \\ 0 \end{bmatrix} \end{bmatrix}$$

7. Obtain the gradient of matrix $f = \begin{bmatrix} \sin(x_0+2x_1) & 2x_1+x_3 \\ 2x_0+x_2 & \cos(2x_2+x_3) \end{bmatrix}$ wrt matrix $x = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$

Sol^n: $$\frac{df}{dx} = \begin{bmatrix} \frac{df_{11}}{dx} & \frac{df_{12}}{dx} \\ \frac{df_{21}}{dx} & \frac{df_{22}}{dx} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} \frac{\partial f_{11}}{\partial x_0} & \frac{\partial f_{11}}{\partial x_1} \\ \frac{\partial f_{11}}{\partial x_2} & \frac{\partial f_{11}}{\partial x_3} \end{bmatrix} & \begin{bmatrix} \frac{\partial f_{12}}{\partial x_0} & \frac{\partial f_{12}}{\partial x_1} \\ \frac{\partial f_{12}}{\partial x_2} & \frac{\partial f_{12}}{\partial x_3} \end{bmatrix} \\ \begin{bmatrix} \frac{\partial f_{21}}{\partial x_0} & \frac{\partial f_{21}}{\partial x_1} \\ \frac{\partial f_{21}}{\partial x_2} & \frac{\partial f_{21}}{\partial x_3} \end{bmatrix} & \begin{bmatrix} \frac{\partial f_{22}}{\partial x_0} & \frac{\partial f_{22}}{\partial x_1} \\ \frac{\partial f_{22}}{\partial x_2} & \frac{\partial f_{22}}{\partial x_3} \end{bmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) \\ 0 & 0 \end{bmatrix}_{2\times2} & \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix}_{2\times2} \\ \begin{bmatrix} 2 & 0 \\ 1 & 0 \end{bmatrix}_{2\times2} & \begin{bmatrix} 0 & 0 \\ -2\sin(2x_2+x_3) & -\sin(2x_2+x_3) \end{bmatrix}_{2\times2} \end{bmatrix}_{2\times2}$$

8. Find the derivative of the matrix $f = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ wrt $x = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$

Sol^n: $$\frac{df}{dx} = \begin{bmatrix} \frac{df_{11}}{dx} & \frac{df_{12}}{dx} \\ \frac{df_{21}}{dx} & \frac{df_{22}}{dx} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 2x_0x_1x_2 & x_0^2 x_2 \\ x_0^2 x_1 & 0 \end{bmatrix} & \begin{bmatrix} 0 & 2x_1^2x_2x_3 \\ x_1^2 x_3 & x_1^2 x_2 \end{bmatrix} \\ \begin{bmatrix} 0 & x_3^2 x_3 \\ 0 & 3x_3^2 x_1 \end{bmatrix} & \begin{bmatrix} 0 & x_2^3 \\ 3x_2^2 & 0 \end{bmatrix} \end{bmatrix}$$

9. Obtain the gradient of matrix $\begin{bmatrix} xyz & xy \\ zy & zx \\ z & 2yx \end{bmatrix}$ wrt $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\frac{df}{dx} = \begin{bmatrix} \frac{df_{11}}{dx} & \frac{df_{12}}{dx} \\ \frac{df_{21}}{dx} & \frac{df_{22}}{dx} \\ \frac{df_{31}}{dx} & \frac{df_{32}}{dx} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} yz \\ xz \\ xy \end{bmatrix} & \begin{bmatrix} y \\ x \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ z \\ y \end{bmatrix} & \begin{bmatrix} z \\ 0 \\ x \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 2y \\ 2x \\ 0 \end{bmatrix} \end{bmatrix}$$

10. obtain the gradient of the matrix $\begin{bmatrix} xy & yz & zw \\ wx & wy & wz \end{bmatrix}$ wrt $\begin{bmatrix} x & y \\ z & w \end{bmatrix}$

$$\frac{df}{dx} = \begin{bmatrix} \frac{df_{11}}{dx} & \frac{df_{12}}{dx} & \frac{df_{13}}{dx} \\ \frac{df_{21}}{dx} & \frac{df_{22}}{dx} & \frac{df_{23}}{dx} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} y & x \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 0 & z \\ y & 0 \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ w & z \end{bmatrix} \\ \begin{bmatrix} w & 0 \\ 0 & x \end{bmatrix} & \begin{bmatrix} 0 & w \\ 0 & y \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ w & z \end{bmatrix} \end{bmatrix}.$$

2 Given $f = \begin{bmatrix} \sin(x_0+2x_1) & 2x_1+x_3 \\ 2x_0+x_2 & \cos(2x_2+x_3) \end{bmatrix}$ & $X = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$ verify

i) $\dfrac{\partial(\text{Trace}(f(x)))}{\partial X} = \text{Trace}\left(\dfrac{\partial f(X)}{\partial X}\right)$

$\text{Trace } f(X) = \sin(x_0+2x_1) + \cos(2x_2+x_3)$

$$\therefore \frac{\partial(\text{Trace } f(x))}{\partial X} = \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) \\ -2\sin(2x_2+x_3) & -2\sin(2x_2+x_3) \end{bmatrix} \quad -(1)$$

$$\text{Trace}\left(\frac{\partial f(x)}{\partial X}\right) = \frac{\partial}{\partial X}\sin(x_0+2x_1) + \frac{\partial}{\partial X}\cos(2x_2+x_3)$$

$$= \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -2\sin(2x_2+x_3) & -\sin(2x_2+x_3) \end{bmatrix}$$

$$= \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) \\ -2\sin(2x_2+x_3) & -2\sin(2x_2+x_3) \end{bmatrix} \quad -(2)$$

From (1) & (2) $\dfrac{\partial(\text{Trace } f(x))}{\partial X} = \text{Trace}\left(\dfrac{\partial f(x)}{\partial X}\right)$

ii) $\dfrac{\partial f(x)^T}{\partial X} = \left(\dfrac{\partial f(x)}{\partial X}\right)^T$

$$f(x)^T = \begin{bmatrix} \sin(x_0+2x_1) & 2x_0+x_2 \\ 2x_1+x_3 & \cos(2x_2+x_3) \end{bmatrix}$$

$$\frac{\partial (f(x))^T}{\partial X} = \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & -2\sin(2x_2+x_3) & -\sin(2x_2+x_3) \end{bmatrix} \quad -(1)$$

$$\frac{\partial f(x)}{\partial X} = \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & 0 & 0 \\ 1 & 0 & -2\sin(2x_2+x_3) & -\sin(2x_2+x_3) \end{bmatrix}$$

$$\left(\frac{\partial f(x)}{\partial x}\right)^T = \begin{bmatrix} \cos(x_0+2x_1) & 2\cos(x_0+2x_1) & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & -2\sin(2x_2+x_3) & -\sin(2x_2+x_3) \end{bmatrix} - (2)$$

From (1) & (2) $\frac{\partial f(x)^T}{\partial x} = \left(\frac{\partial f(x)}{\partial x}\right)^T$

3. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ $a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ & $b = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ verify $\frac{\partial (a^T x b)}{\partial x} = ab^T$

Soln: $a^T x b = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$= -x_1 + x_3 + x_2 - x_4$

$= -x_1 + x_2 + x_3 - x_4$

$\frac{\partial (a^T x b)}{\partial x} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - (1)$

$ab^T = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} - (2)$

From (1) & (2) $\frac{\partial}{\partial x}(a^T x b) = ab^T$

4. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, $a = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ & $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ verify $\frac{\partial (a^T x b)}{\partial x} = ab^T$

Soln: $a^T x b = \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2x_1 + 3x_3 + 4x_2 + 6x_4$

$\frac{\partial (a^T x b)}{\partial x} = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} - (1)$

$ab^T = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} - (2)$

From (1) & (2) $\frac{\partial}{\partial x}(a^T x b) = ab^T$

5. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ & $a = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ verify $\frac{\partial}{\partial x}(x^T a) = a^T$

Soln: $x^T = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = 2x_1 - x_2 + 4x_3$

$\frac{\partial}{\partial x}(x^T a) = \cancel{\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}} \begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$

$a^T = \begin{bmatrix} 2 & -1 & 4 \end{bmatrix}$

6. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ and $a = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$ verify $\frac{\partial (a^T x)}{\partial x} = a^T$

Soln: $a^T x = \begin{bmatrix} 1 & 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_1 + 2x_2 + 3x_3 + x_4$

$\partial/\partial x \, (a^T x) = \begin{bmatrix} 1 & 2 & 3 & 1 \end{bmatrix}$ — (i)

$a^T = \begin{bmatrix} 1 & 2 & 3 & 1 \end{bmatrix}$ — (ii)

$\therefore \frac{\partial}{\partial x}(a^T x) = a^T$

Convex Sets : A set C is convex if the line segment between any two points in C lies in C.

i.e, a set $C \subseteq \mathbb{R}^n$ is convex, if for all $x, y \in C$ & $\forall \lambda \in [0,1]$

$\lambda x + (1-\lambda) y \in C$.

A point of the form $\lambda x + (1-\lambda) y$, $\lambda \in [0,1]$ is called a convex combination of x & y. As λ varies b/w $[0,1]$, a line segment is being formed b/w x & y.

Ex : point, Triangle, circle, ellipse.

nonconvex – Torus, ring, star, Cardioide

1 Show that intersection of two convex sets is a convex set. Is union of two convex sets is necessarily convex set: Justify your answer.

Let C_1 & C_2 be convex sets. Suppose $x, y \in C_1 \cap C_2$.

Since $x, y \in C_1$ and C_1 is convex. then $\lambda x + (1-\lambda) y \in C_1$

similary $x, y \in C_2$ & C_2 is also convex then $\lambda x + (1-\lambda) y \in C_2$

$\therefore$ The combination is in both sets, so $\lambda x + (1-\lambda) y \in C_1 \cap C_2$

Hence intersection of two convex sets is a convex set.

No, the union of two convex sets is not necessarily convex.

Consider two disjoint points in $\mathbb{R}$

Let $C_1 = \{1\}$ & $C_2 = \{3\}$ are two convex sets. Their union is $C_1 \cup C_2 = \{1, 3\}$. The line segment between them includes 2, but $2 \notin \{1, 3\} = C_1 \cup C_2$.

2. Show that the unit ball $B = \{u \in \mathbb{R}^n : \|u\| < 1\}$ is convex set

Let $x, y \in B$. i.e., $\|x\| < 1$ & $\|y\| < 1$.

For any $\lambda \in [0, 1]$, Let $z = \lambda x + (1-\lambda) y$.

using the triangle inequality

$$\|z\| = \|\lambda x + (1-\lambda) y\| \leq \lambda \|x\| + (1-\lambda) \|y\|$$

Since $\|x\| < 1$ & $\|y\| < 1$

$$\therefore \|z\| < \lambda + 1 - \lambda = 1 \Rightarrow \|z\| < 1.$$

Hence $z \in B$. The set is convex

3. Show that $S = \{(x_1, x_2) : 2x_1 + 3x_2 = 7\} \in \mathbb{R}^2$ is a convex set

Let $x, y \in S$: Then $2x_1 + 3x_2 = 7$ & $2y_1 + 3y_2 = 7$

~~For $z = \lambda x + (1-\lambda) y$~~ Now consider a point z on the line segment between x & y, where $z = \lambda x + (1-\lambda) y$ for $\lambda \in [0, 1]$.

The coordinates of z are:

$$z = (\lambda x_1 + (1-\lambda) y_1, \lambda x_2 + (1-\lambda) y_2)$$

Substitute these into the set S.

$$2(\lambda x_1 + (1-\lambda) y_1) + 3(\lambda x_2 + (1-\lambda) y_2) \ \text{eq}$$

$$= \lambda (2x_1 + 3x_2) + (1-\lambda)(2y_1 + 3y_2) \ \text{eq}$$

$$z\lambda + (1-\lambda) z = z$$

Since the equation holds, $z \in S$.

Thus S is a convex set.

4. Show that $S = \{(x_1, x_2) : x_1^2 + x_2^2 \le 4\}$ is a convex set.

This set represents a disk with radius 2 centered at the origin. Let $x, y \in S$, so $\|x\| \le 2$ & $\|y\| \le 2$. → radius

For $\lambda \in [0,1]$, let $z = \lambda x + (1-\lambda) y$.

By using triangle inequality & properties of norms

$$\|z\| = \|\lambda x + (1-\lambda) y\| \le \lambda \|x\| + (1-\lambda)\|y\|$$

$$\Rightarrow \|z\| \le 2\lambda + 2(1-\lambda)$$

$\Rightarrow \|z\| \le 2$, then $x_1^2 + x_2^2 \le 4$ holds for z.

Thus S is a convex set

unit ball in $\mathbb{R}^3$

5. Show that $S = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 + x_3^2 \le 1\}$ is a convex set

Let $x, y \in S$, $\|x\| \le 1$ & $\|y\| \le 1$.

By the def^n of convex set $z = \lambda x + (1-\lambda) y$

Using triangle inequality

$$\|z\| = \|\lambda x + (1-\lambda) y\| \le \lambda \|x\| + (1-\lambda)\|y\|$$

$$\Rightarrow \|z\| \le \lambda + 1 - \lambda$$

$\Rightarrow \|z\| \le 1$, then $x_1^2 + x_2^2 + x_3^2 \le 1$ holds for z.

Thus S is a convex set

6. Show that $S = \{(x_1, x_2, x_3) : 2x_1 - x_2 + x_3 \le 4\}$ is a convex set

This set represent a half-space.

Let $x, y \in S$, then $2x_1 - x_2 + x_3 \le 4$ & $2y_1 - y_2 + y_3 \le 4$.

$z = \lambda x + (1-\lambda) y$ for $\lambda \in [0,1]$

$\therefore$ The coordinates of z are

$$z = (\lambda x_1 + (1-\lambda) y_1, \ \lambda x_2 + (1-\lambda) y_2, \ \lambda x_3 + (1-\lambda) y_3)$$

Substitute this in set S.

$$2(\lambda x_1 + (1-\lambda) y_1) - (\lambda x_2 + (1-\lambda) y_2) + (x_3\lambda + (1-\lambda) y_3)$$

$$= \lambda(2x_1 - x_2 + x_3) + (1-\lambda)(2y_1 - y_2 + y_3)$$

$$\le 4\lambda + (1-\lambda) 4$$

$z \le 4$, $z \in S$. Thus S is a convex set

Hyperplane : A hyperplane is a flat affine subspace of one dimension less than its ambient space. In an n-dimensional space, a hyperplane has $(n-1)$ dimensions.

For given scalar d in $\mathbb{R}$, the set of all $(x_1, x_2, \dots x_n) \in \mathbb{R}^n$ & $a_1x_1 + a_2x_2 + \dots + a_nx_n = d$ $(f(x) = d)$ is called hyperplane

i.e, if $n = (a_1, a_2 \dots a_n)$ & d is scalar then hyperplane is the set $H = \{X \in \mathbb{R}^n : n \cdot X = d\}$. n is normal vector.

Def^n : The hyperplane $H = [f : d]$ separates two sets A & B if one of the following holds :

$[f : d]$ denotes the set of all x in $\mathbb{R}^n$ at which the value of f is d

i) $f(A) \le d$ & $f(B) \ge d$ (or)

ii) $f(A) \ge d$ & $f(B) \le d$

Ex : In $\mathbb{R}^2$, the line $x - 4y = 13$ is a

In the conditions above, if all the weak inequalities are replaced by strict inequalities, then H is said to strictly separate A & B.

Properties of Hyperplane :

* A hyperplane divides the space into two half-spaces.
* All points on one side of the hyperplane satisfy $a_1x_1 + a_2x_2 + \dots + a_nx_n > d$, while those on the other side satisfy $a_1x_1 + a_2x_2 + \dots + a_nx_n < d$

① Let $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ & $v = \begin{bmatrix} 1 \\ -6 \end{bmatrix}$ & let $H = \{X : A \cdot X = 12\}$. Find the implicit description of the parallel hyperplane $H_1 = H + v$.

Given $H = \{X : A \cdot X = 12\}$, $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ -6 \end{bmatrix}$

we have $A \cdot X = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 \end{bmatrix} = 3x_1 + 4x_2$.

$\therefore A \cdot X = 12 \Rightarrow 3x_1 + 4x_2 = 12$ $\quad H = [f : 12]$

Here the function $f(x_1, x_2) = 3x_1 + 4x_2$ has the value 12.

The hyperplane H_1 is translated by the vector v

so $H_1 = H + v$.

If $y \in H_1$ then $y = x + v$ for some $x \in H$.

$\Rightarrow x = y - v$

Substitute x in H.

$$A \cdot (Y - v) = 12$$
$$\Rightarrow A \cdot Y - A \cdot V = 12$$
$$\Rightarrow A \cdot Y = 12 + A \cdot V \quad \text{— ①}$$

$$A \cdot V = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -6 \end{bmatrix} = (3 \times 1) + (4 \times -6) = 3 - 24 = -21$$

$\therefore$ eq^n ① becomes $A \cdot Y = 12 - 21 = -9$

The implicit description of the hyperplane H, is

$$3x_1 + 4x_2 = -9 \text{ i.e, } H = [f : -9]$$

2. Let $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ & $v_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1 & v_2.

A hyper plane in $\mathbb{R}^2$ is defined by the eq^n : $ax_1 + bx_2 = d$ where $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is a point on the hyperplane & $n = \begin{bmatrix} a \\ b \end{bmatrix}$ is the normal vector.

Since H passes through v_1 & v_2 points must satisfy the eq^n

For v_1 : $a + 3b = d$ — (i) $\quad n \cdot v_1 = d$

For v_2 : $4a = d$ — (ii) $\quad n \cdot v_2 = d$

Equate two equations $\quad 4a = a + 3b$

$$\Rightarrow 3a = 3b$$
$$\Rightarrow \boxed{a = b}$$

Substitute $a = b$ in (i), we get $d = 4a$.

Let $a = b = 1$, then $d = 4$.

Thus the implicit equation is $\boxed{x_1 + x_2 = 4}$

3. Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ $v_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$ & $v_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1, v_2 & v_3.

Sol^n: A hyperplane in $\mathbb{R}^3$ is defined by $ax_1 + bx_2 + cx_3 = d$ —(*)

where $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ & $n = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

Since H passes through v_1, v_2 & v_3

3. Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$.

Soln: $d_1 = v_2 - v_1 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$

$d_2 = v_3 - v_1 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$

The normal vector n to the hyperplane is perpendicular to both direction vectors. We find it using the cross product.

$n = d_1 \times d_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -2 \\ -2 & -3 & 2 \end{vmatrix} = \hat{i}(6-6) - \hat{j}(2-4) + \hat{k}(-3+6)$

$= 0\hat{i} + 2\hat{j} + 3\hat{k}$

$\therefore$ The normal vector is $n = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$

The implicit eqn is $n \cdot x = d \Rightarrow 2x_2 + 3x_3 = d$ — (1)

Substitute v_1 in (1), $d = 11$

Thus the hyperplane H is described by the eqn $2x_2 + 3x_3 = 11$.

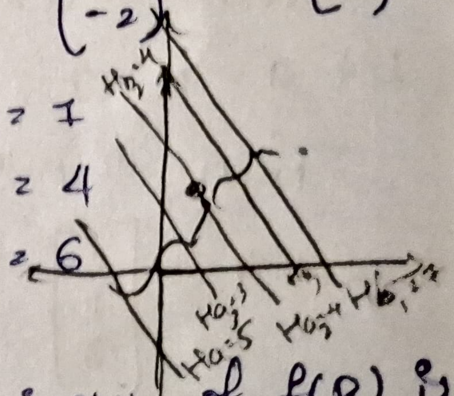
To find d : from (i), $2+9=d \Rightarrow d=11$

Thus the implicit eq^n of the hyperplane is

$$0x_1 + 2x_2 + 3x_3 = 11 \Rightarrow \boxed{2x_2 + 3x_3 = 11}$$

4 Let $A = \left\{ \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix} \right\}$, $B = \left\{ \begin{bmatrix} 0 \\ 5 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$ & $n = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$. Find a hyperplane H with normal n that separates A & B. Is there any hyperplane that separates A & B strictly?

Sol^n: Let $a_1 = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$, $a_2 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$, $a_3 = \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix}$, $b_1 = \begin{bmatrix} 0 \\ 5 \\ -1 \end{bmatrix}$ $b_2 = \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}$ $b_3 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$



Now $n \cdot a_1 = 6 - 1 - 10 = -5$

$n \cdot a_2 = 9 + 1 - 6 = 4$

$n \cdot a_3 = -3 + 6 + 0 = 3$

$\therefore A \in [-5, 4]$

$n \cdot b_1 = 0 + 5 + 2 = 7$

$n \cdot b_2 = 3 - 3 + 4 = 4$

$n \cdot b_3 = 6 + 2 - 2 = 6$

$\therefore B \in [4, 7]$

Since the maximum of $f(A)$ is 4 and the minimum of $f(B)$ is 4, they share the value 4.

$\therefore H = 3x_1 + x_2 - 2x_3 = 4$.

No, strict separation is possible with this norm because both sets contain a point that result in exactly 4, thus touches the hyperplane.

5. Let $A = \left\{ \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 2 \end{bmatrix} \right\}$, $B = \left\{ \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\}$ and $n = \begin{bmatrix} -3 \\ 2 \\ 2 \end{bmatrix}$.

Find a hyperplane H with normal n separates A & B. Is there any hyperplane that separates A & B strictly?

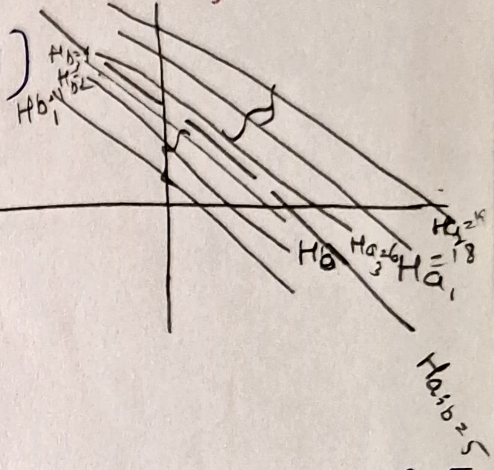
Soln: By the defn of $H = \{ x \in \mathbb{R}^n : n \cdot x = d \}$

$n \cdot a_1 = 6 + 2 + 10 = 18$

$n \cdot a_2 = -6 + 2 + 10 = 6$

$n \cdot a_3 = 3 + 12 + 4 = 19$

$\therefore A \in [6, 19]$



$n \cdot b_1 = 8 - 4 = 4$

$n \cdot b_2 = 3 - 6 + 4 = 1$

$n \cdot b_3 = -6 + 4 + 4 = 2$

$\therefore B = [1, 4]$

Since all values of $n^T b$ are strictly less than all values of $n^T a$ any d in the interval $[4, 6)$ defines a separating hyperplane. Choose $d = 5$, the hyperplane H is

$$H : -3x_1 + 2x_2 + 2x_3 = 5.$$

Yes. Two sets can be strictly separated if their convex hulls are disjoint.
As calculated above, the values of $n^T x$ for sets A & B do not overlap: $\min_{a \in A} (n^T a) = 6$, $\max_{b \in B} (n^T b) = 4$
Because $4 < 6$, any d chosen strictly between 4 & 6 results in a strict separation, where $n^T b < d < n^T a$.